Nutrition Coaching Calculation Logic (v2)

Calorie targets, macro distribution, and meal-splitting rules — updated draft with corrected BMR coefficient, full heavy-variant tables, and database-driven macro-per-meal method

1. Client & Goal

Every client has a single goal, which determines how the target calories are set relative to maintenance:

- Lose weight
- Maintain weight
- Gain weight

Verdict

Good: Clear and standard three-way split. No issues here.

Missing: No body-recomposition case (simultaneous fat loss + muscle gain), which is common for beginners or clients returning to training. Not required, but worth noting as a future extension.

2. BMR Calculation — Mifflin-St Jeor Equation

Note: the correct name is the Mifflin-St Jeor equation (not "Mifli-Sanjon" — likely a mishearing). Confirmed against current sources; using the correct name in the final report avoids confusion if it's ever shared with another dietitian.

Inputs: weight (kg), height (cm), age (years), sex.

Men

BMR = 10 × weight + 6.25 × height − 5 × age + 5

Women

BMR = 10 × weight + 6.25 × height − 5 × age − 161

Correction from previous draft: the height coefficient is 6.25 for both sexes. The only difference between the two equations is the final constant (+5 for men, −161 for women).

Verdict

Good: Both equations now match the published Mifflin-St Jeor formula exactly.

Note: Mifflin-St Jeor is well-supported in the literature as more accurate than Harris-Benedict for most populations, though it was validated primarily on a Western/American sample and can be less precise for very muscular, very old, or clinically obese clients.

3. Activity Level → TDEE (Maintenance Calories)

Maintenance calories = BMR × activity factor:

Activity level	Factor	Definition
Sedentary	1.2	Little to no gym or training
Light activity	1.375	3–4 activity sessions per week
Moderate	1.55	4–5 training sessions per week
Athlete	1.9	Physically demanding job/lifestyle

Verdict

Good: These four factors are the standard, widely-used TDEE multipliers.

Problem: 1.9 for "athlete" is on the extreme end — commonly reserved for elite/professional athletes in heavy daily training, not just "a physically demanding job." A physical job (construction, etc.) without structured training usually sits closer to 1.6–1.7. Using 1.9 for a physical-job client risks overestimating maintenance calories significantly, which would sabotage a fat-loss goal.

Improve: Consider splitting "physical job" and "athlete/very intense training" into two separate tiers rather than merging them, since they represent meaningfully different energy expenditures.

4. Adjusting Maintenance for the Goal

Maintain

Target calories = Maintenance calories (no change).

Lose Weight

Target weekly loss: 0.5%–1% of body weight per week.

- 1 kg of body weight ≈ 7,700 kcal (energy equivalent used for the calculation).
- Weekly deficit (kcal) = (body weight × chosen % ÷ 100) × 7,700

- Daily deficit (kcal) = Weekly deficit ÷ 7
- Daily target calories = Maintenance − Daily deficit

Example: 100 kg client, targeting 1% loss/week → 1 kg/week → 7,700 kcal/week deficit → 1,100 kcal/day deficit.

Gain Weight

Target calories = Maintenance + 200 to 300 kcal (fixed surplus, not weight-scaled).

Verdict

Good: The 0.5–1% bodyweight/week loss rate is a well-supported, sustainable range used broadly in evidence-based coaching. The 7,700 kcal/kg constant is the commonly cited (if approximate) energy density of body fat/adipose tissue changes, so it's an acceptable simplification for coaching purposes.

Problem: At high body weights (e.g. 150 kg client, 1%/week), the resulting daily deficit can be very large and may push target calories below a safe floor (risk of falling under BMR or under ~1200 kcal for many clients). The document has no minimum-calorie safety floor.

Improve: Add a hard floor rule, e.g. "target calories must never fall below 1.2× BMR, or an absolute minimum (commonly ~1200–1500 kcal depending on sex/size), whichever is more conservative." Also, the gain-weight surplus (200–300 kcal) is flat regardless of body size — reasonable for most clients, but very large or very small individuals might warrant scaling it too.

5. Macronutrient Distribution

Order of operations: protein is set first, fat second, and carbohydrates fill the remaining calories.

Macro	Range (g/kg body weight)	Recommended default	kcal per gram
Protein	1.8 – 2.2	2.0	4
Fat	0.66 – 1.0	0.7	9
Carbohydrate	Remaining calories ÷ 4	—	4

Worked example — 100 kg client, target 2,500 kcal/day:

- Protein: 100 × 2.0 = 200 g → 800 kcal
- Fat: 100 × 0.7 = 70 g → 630 kcal

- Remaining: 2,500 − 800 − 630 = 1,070 kcal
- Carbs: 1,070 ÷ 4 = 267.5 g

Verdict

Good: Protein and fat ranges are both within widely-supported evidence-based bounds for active individuals (protein especially — 1.8–2.2 g/kg aligns with sports-nutrition literature for muscle retention during a deficit). Filling remaining calories with carbs is the standard, simplest approach.

Problem: Protein and fat targets are calculated from total body weight, not lean body mass. For clients with higher body-fat percentages, this meaningfully overshoots true protein/fat needs (fat tissue doesn't need 2g/kg protein support the way muscle does). This is the single biggest scientific loophole in the whole document.

Improve: Consider switching to lean body mass (or an estimated fat-free mass) for the protein/fat calculation once you have a way to estimate body composition — even a rough visual/skinfold estimate would improve accuracy for higher body-fat clients.

Edge case: No lower bound is defined for carbs. In an aggressive deficit with a heavy client and high protein/fat targets, the remaining calories for carbs could go very low or even negative — the document needs a rule for what happens if carbs "run out" (e.g. re-balance by trimming fat toward its lower bound first).

6. Calorie Distribution Across Meals

The total daily calories are split across meals as a percentage. Three named patterns exist: Balanced, and three "heavy" variants (breakfast-heavy, lunch-heavy, dinner-heavy).

6.1 Balanced Distribution

# of meals	Distribution
2 meals	40% / 60%
3 meals	25% / 40% / 35%
4 meals	25% / 15% (snack) / 30% / 30%
5 meals	20% / 10% / 30% / 10% / 30%

⚠ Unresolved naming problem (flagged as requested): with 2 meals and with 5 meals, it is not defined which slot corresponds to breakfast, lunch, or dinner. This matters because the "heavy" variants below are defined in terms of increasing a specific named meal (e.g. "dinner-heavy"), but if meal identity isn't fixed per meal-count, it's ambiguous which percentage to boost.

6.2 Breakfast-Heavy Distribution

# of meals	Distribution
2 meals	60% / 40%
3 meals	40% / 30% / 30%

4 meals	35% / 15% (snack) / 25% / 25%
5 meals	30% / 15% / 25% / 10% / 20%

6.3 Lunch-Heavy Distribution

# of meals	Distribution
2 meals	35% / 65% — treat 2nd slot as the lunch/main meal
3 meals	20% / 50% / 30%
4 meals	20% / 10% (snack) / 45% / 25%
5 meals	15% / 10% / 45% / 10% / 20%

6.4 Dinner-Heavy Distribution

# of meals	Distribution
2 meals	30% / 70%
3 meals	20% / 30% / 50%
4 meals	20% / 10% (snack) / 25% / 45%
5 meals	15% / 10% / 20% / 10% / 45%

Rule used to generate these: the target meal's percentage is boosted by roughly 10–20 percentage points versus the balanced version, and the difference is pulled proportionally from the other non-snack meals, while snack slots (the 15%/10% "light" entries in the 4- and 5-meal rows) are left untouched so a snack never becomes unreasonably small or large.

Verdict

Good: Percentage-based meal splitting is a practical, easy-to-apply coaching tool, and having named variants (breakfast/lunch/dinner-heavy) is a nice personalization touch for clients with different schedules (e.g. night-shift workers, early risers).

Problem — flagged previously: The 2-meal and 5-meal cases still don't have hard-assigned meal labels (breakfast/lunch/dinner) — the tables above make a reasonable assumption (2 meals = first meal / main meal; 5 meals = breakfast, snack, lunch, snack, dinner) but this assumption should be confirmed and written down as policy, not left implicit.

Improve: These exact percentages are illustrative — pick one consistent rule (e.g. "+15 points to the target meal, proportionally removed from others, snacks untouched") and regenerate all three tables from that single rule so future edits stay consistent rather than hand-tuned per table.

7. Macro Distribution Across Meals — Database-Driven Ratios

Section 6 only distributes total calories across meals (e.g. 40% / 60%). It does not define how the protein/fat/carb grams themselves are split across those same meals. Rather than picking an arbitrary rule, the approach is:

- Look at existing nutrition databases (and our own client data, once we have enough of it) to find the typical/average macro ratio people actually eat at each meal type — breakfast, lunch, dinner, snack.
- From that average, define either a fixed ratio or an allowed range of ratios (protein : fat : carb) for each meal type.
- When building a client's plan, each meal's calorie allocation (from Section 6) is then split into grams using that meal's typical ratio, instead of just re-using the day's overall calorie percentage for every macro.

Illustrative example only — not real data yet:

- Breakfast: hypothetically found to average roughly an even split across protein, fat, and carbs (~1:1:1 by calorie share) — so if breakfast = 500 kcal, each macro gets ~167 kcal worth, converted to grams via 4/9/4 kcal-per-gram.
- Dinner: hypothetically found to skew higher in protein and lower in fat relative to breakfast.

These numbers are placeholders to illustrate the method — the actual ratios must come from real database/food-log data before this is used with clients.

Verdict

Good: This is a stronger approach than picking an arbitrary split. Published dietary-intake data does show real, consistent differences in how people naturally distribute macros across meals — for example, breakfast tends to skew carbohydrate/fat-heavy and lighter in protein, while dinner (in "even" distribution studies) tends to carry a meaningfully higher protein share. Grounding your ratios in that kind of data is more defensible than an arbitrary rule.

Caution: "Average ratio in the database" reflects what people typically eat, which is not automatically what's optimal. The same research shows that concentrating protein at dinner and skimping at breakfast (a common natural pattern) is generally considered worse for muscle protein synthesis than spreading protein evenly across meals. If you copy the database's average pattern uncritically, you may be encoding a bad habit (protein-skewed dinners, protein-light breakfasts) rather than a good one.

Improve: Use the database ratios as a flavor/palatability reference (what feels normal to eat at each meal) but overlay a protein floor per meal — e.g. don't let any single meal's protein ratio fall so low that daily protein synthesis is compromised, even if that's what the "average" person does at that meal.

Improve: Define whether the ratio is a single fixed number or a range per meal type — a range is more flexible for the ±5% swap logic in Section 8, since it gives the algorithm room to move without leaving the meal's "realistic" macro profile.

8. Flexibility Window for Meal Swaps

Once the initial full-day distribution (calories + macros, across meals) is set, some flexibility is allowed when substituting/adjusting individual meals, so the plan isn't overly rigid.

Rule as defined

- The allowed window is ±5% of total daily calories, applied at the level of each individual meal — i.e. each meal's calories are allowed to move up or down by up to 5% of the day's total calorie target (not 5% of that meal's own calories).
- Within that shift, each macro (protein, fat, carbs) must still fall within its original per-kg range (e.g. protein factor must stay between 1.8–2.2 g/kg, fat between 0.66–1.0 g/kg).

On "isn't per-meal ±5% the same as per-day ±5%?"

Not quite — they bound the same worst case, but they are not equivalent in general:

- If every meal is allowed to swing up to ±5% of the day's total, and every meal happens to swing in the same direction at once, the whole day can drift by up to ±5% — so per-meal ±5% (of the daily total) can never produce a daily drift larger than 5%.
- But in the typical case, meals don't all swing the same way — one meal runs a bit high while another runs a bit low — and those differences partially cancel out. In that case the day-level drift ends up smaller than 5%, even though each individual meal used its full ±5% allowance.
- So constraining each meal individually guarantees the day stays within the window, but the day usually won't actually reach the full ±5% — it's a safe, slightly conservative way to bound the total, not a guarantee that the total always hits exactly ±5%.

Verdict

Good: Defining the ±5% as a share of total daily calories (not of each meal's own calories) is the right call — it means a small snack and a large dinner get the same absolute flexibility in kcal, rather than the snack having almost no room to move.

Good: This is a sound way to give a nutrition plan meal-swap flexibility without letting it drift from the client's evidence-based targets — it mirrors how real coaching apps handle "close enough" substitutions.

Confirmed: Applying ±5% of the daily total to each meal independently is a safe (if occasionally slightly conservative) way to bound the whole day within ±5% — worst case they all move the same direction and the day hits the full 5%; typical case some cancel out and the day moves less than 5%. This is a reasonable, low-risk rule to keep.

Problem: The macro-range constraint (1.8–2.2 g/kg protein, etc.) is a whole-day target, but it's being applied here as if it were a per-meal constraint. A single meal's protein content swinging within the *daily* g/kg range doesn't translate directly to that one meal — this needs either a per-meal version of the range (e.g. derived from that meal's share of daily calories combined with Section 7's per-meal ratios), or the constraint should be re-framed as "the daily total, after all meal adjustments, must still fall in range," checked only after all swaps are applied rather than per meal.

Improve: Recommend checking the macro constraint at the daily-total level after swaps are applied, rather than meal-by-meal — it's simpler to implement, and it's more forgiving for small meals/snacks where a per-meal g/kg range is not really meaningful.

9. Overall Summary Verdict

Where the framework is strong:

- The overall pipeline (BMR → TDEE → goal adjustment → macros → meal split → meal-swap flexibility) is a logical, complete structure that mirrors how professional coaching apps are actually built.

- The core numeric ranges (protein 1.8–2.2 g/kg, fat 0.66–1.0 g/kg, 0.5–1%/week weight loss rate) are all within accepted evidence-based bounds.

Fixed since the previous draft:

- Men's BMR height coefficient corrected to 6.25 (matches the published equation).
- Breakfast-heavy, lunch-heavy, and dinner-heavy tables now have explicit numbers for 2/3/4/5-meal counts, generated from one consistent boost-and-redistribute rule.
- Macro-per-meal distribution now has a concrete method (database-derived per-meal ratios) instead of being an open question.
- The ±5% flexibility window is now explicitly defined as ±5% of daily total calories, applied per meal — and the "isn't this the same as per-day?" question is answered (it bounds the same worst case but isn't strictly equivalent).

Still needs work before being usable end-to-end:

- Resolve the athlete/physical-job multiplier — likely split into two tiers (physical job vs. elite athlete).
- Add a minimum safe-calorie floor for the lose-weight case.
- Confirm the assumed meal labels for the 2-meal and 5-meal cases (documented as an assumption in Section 6, not yet formally confirmed as policy).
- Source real per-meal macro ratio data (Section 7 currently uses illustrative placeholder numbers, not actual database output) and decide on a protein floor per meal so the database averages don't just replicate a protein-light-breakfast pattern.
- Decide whether the macro-range check in Section 8 happens per meal or only on the daily total after all swaps (recommended: daily total).
- Consider using lean body mass rather than total body weight for protein/fat targets, for higher body-fat clients.

None of these are fatal flaws — they're the normal set of gaps you'd expect in a first draft of a coaching logic spec, and all are fixable without changing the overall architecture.